Schur–Secular Sub-Quarter Contraction Certificates

Full-Reference-Free Validation of Rank-One Packet Correction

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Abstract

RH-89 showed that one complement direction and an $(r+1)$-dimensional Ritz solve recover more than 96% of the full floating correction dividend at all ten anchors. This paper removes the full reference packet from the finite contraction certificate.

Write the enriched compression as

$$H = \begin{pmatrix} A & b \\ b^* & d \end{pmatrix}.$$

Its rank-r correction gain over the old coordinate packet is exactly $\Delta = d - \lambda_{\min}(H)$. For any $\delta \geq 0$ and vector x, define

$$\Phi_\delta(x) = x^*(A - (d-\delta)I)x - 2\,\mathrm{Re}(x^*b) + \delta.$$

The Schur trial gain certificate states that $\Phi_\delta(x) \leq 0$ implies $\Delta \geq \delta$. It is a one-vector Rayleigh test and needs neither an inverse nor an eigengap. If C is the old-packet predictor tail, E the previous memory tail, and $\delta = C - \rho E$, the target contraction corollary gives a corrected tail at most ρE.

For $\rho = 0.24$, one archived channel is already below target before correction. In all nine remaining channels, a binary linear-solve trial has a strictly negative 256-bit Schur form. The smallest actual-to-required gain ratio exceeds 1.003, and direct corrected residuals are below $0.24E$ in all ten channels. The hardest certified Schur margin is about 2.9×10^{-16}; solve dimensions never exceed seven.

This is a full-reference-free certificate for the frozen models, not a uniform Schur-margin theorem. Stage A, Hilbert–Polya, and the Riemann Hypothesis remain open.

Keywords: Schur complement; secular equation; Rayleigh certificate; dynamic packet; validated numerics.

MSC 2020: 47A75; 65F15; 15A18; 65G20.

1 Removing the full reference packet

RH-89 constructs the old-packet/new-Gram cross direction and solves a rank-r Ritz problem in dimension $r+1$ [3]. Its audit compared the small corrected packet with a full floating reference packet. For an all-level proof this comparison is unnecessary: one only needs enough gain to cross a prescribed contraction threshold.

Let H be the Hermitian compression to the enriched coordinates, partitioned as above. The old packet captures $\operatorname{tr} A$. Retaining the top r eigenvalues of H captures $\operatorname{tr} H - \lambda_{\min}(H)$, so the exact gain is

$$\Delta = d - \lambda_{\min}(H). \tag{1}$$

2 Schur trial gain certificate

Theorem 2.1 (Schur trial gain certificate)**.** *For any $\delta \geq 0$ and $x \in \mathbb{C}^r$, if*

$$\Phi_\delta(x) = x^*(A - (d - \delta)I)x - 2\,\mathrm{Re}(x^*b) + \delta \leq 0, \tag{2}$$

then $\Delta \geq \delta$.

Proof. Set $y = (x, -1)$. Direct expansion gives

$$y^*(H - (d - \delta)I)y = \Phi_\delta(x) \leq 0.$$

The Rayleigh principle therefore implies $\lambda_{\min}(H) \leq d - \delta$. Insert this inequality into (1) [1, 2]. □

No inverse is part of the theorem. Numerically, one may choose x by approximately solving $(A-(d-\delta)I)x = b$ and then validate (2) directly. Ill conditioning affects how the trial is found, but not the logic of the final sign certificate.

Corollary 2.2 (Target contraction corollary)**.** *Let $E > 0$ be the previous memory tail and C the new tail obtained by retaining the old packet. Fix $0 < \rho < 1$ and set $\delta = C - \rho E$. If $\delta \leq 0$, the predictor already meets the target. If $\delta > 0$ and* (2) *holds, the rank-one Ritz corrected tail is at most ρE.*

Proof. The corrected tail equals $C - \Delta$. Apply Theorem 2.1 to obtain $C - \Delta \leq C - \delta = \rho E$. □

This criterion needs only one rank-r trial solve and one scalar sign.

3 Ten-channel 256-bit audit

We set $\rho = 0.24$ at the final predictor-corrector update. The old and corrected residual energies are evaluated directly after exact binary lifting. The small Schur form uses the frozen compressed matrix and a lifted floating trial vector. The enriched-basis orthogonality defect remains below 2×10^{-15}.

σ	channels needing correction	min gain ratio	min negative margin	max corrected
0.16	2	1.57	4.0×10^{-8}	0.0043
0.08	2	2.49	1.0×10^{-12}	0.0212
0.04	1	2.09	3.2×10^{-14}	0.0584
0.02	2	1.003	2.9×10^{-16}	0.2385
0.01	2	1.10	4.7×10^{-15}	0.1900

Table 1: Worst rounded Schur and direct-contraction quantities by scale. Gain ratio means actual small Ritz gain divided by the gain required for the 0.24 target.

The near-active case is the left channel at $\sigma = 0.02$. Its trial system is highly conditioned, but the theorem uses only the final quadratic sign, which remains strictly negative at 256 bits. No full optimal packet appears in this certificate.

4 Boundary and next theorem

The remaining analytic statement is now small and explicit: after a suitable burn-in, construct a cross-block trial vector $x_{\sigma,j}$ and prove $\Phi_{C-\rho E}(x_{\sigma,j}) \leq 0$ uniformly with clock rank $O(\log(1/\sigma))$. A proof may estimate the three terms in (2) directly, without tail eigengaps or ambient singular vectors.

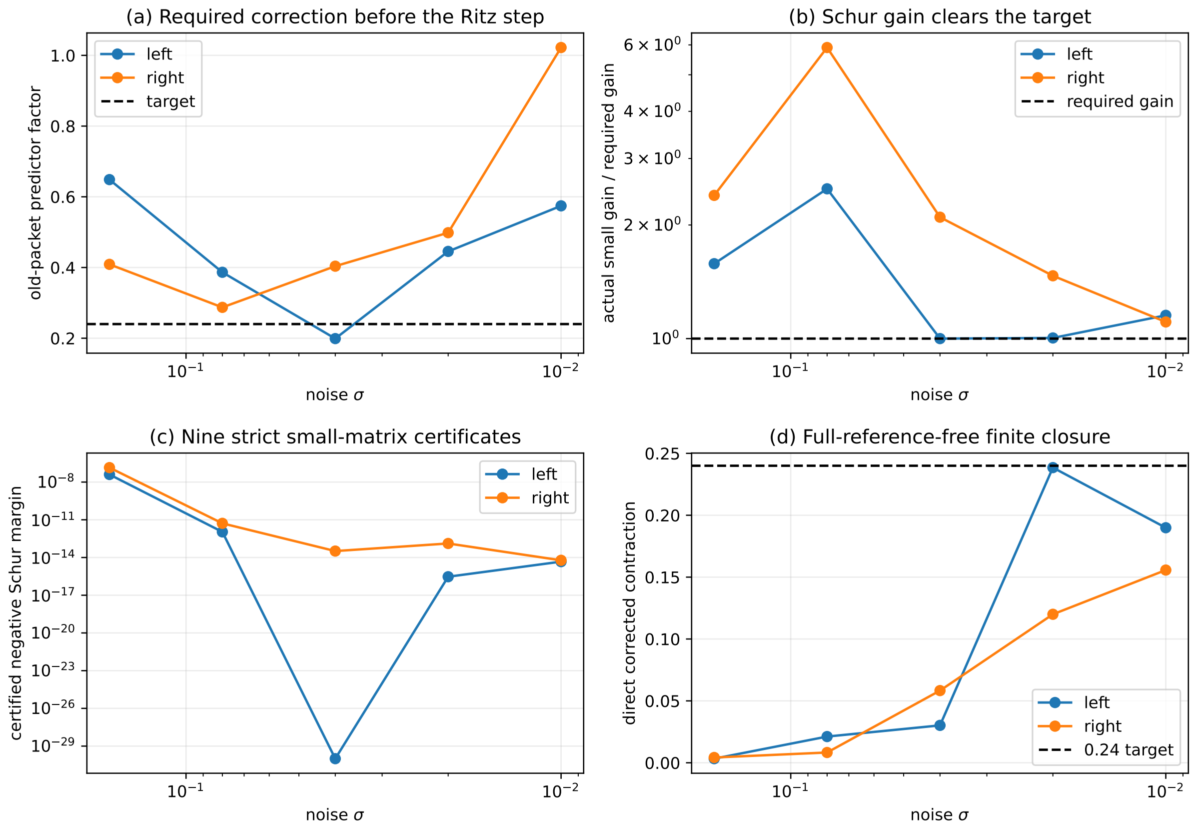

Figure 1: Predictor demand, gain sufficiency, certified Schur margins, and direct full-reference-free contraction.

RH-90 proves the Schur and target-contraction results and validates all ten frozen channels. It does not prove a uniform Schur margin, close Stage A1 or Stage A4, construct a relative determinant or self-adjoint Hilbert–Polya operator, identify zeta zeros, or prove the Riemann Hypothesis.

References

[1] Rajendra Bhatia. *Matrix Analysis*. Springer, 1997.

[2] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, second edition, 2013.

[3] Liang Wang. Rank-one complement ritz correction for dynamic packet energy. 2026. RH-89 manuscript.